

EKIMCHAN, Russian Federation

WMO: 313290

Lat: 53.0749N

Lon: 132.9510E

Elev: 542

StdP: 94.98

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.6	-39.7	-44.1	0.1	-41.3	-42.0	0.1	-39.5	5.7	-13.9	4.2	-20.6	0.5	70	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	28.7	18.0	26.7	17.4	24.8	16.6	19.9	24.9	19.0	23.7	18.1	22.6	2.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.3	21.3	17.5	13.4	20.6	16.5	12.6	19.9	59.8	24.9	56.7	23.7	53.6	22.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.2	4.3	-44.0	31.1	2.1	1.6	-45.5	32.3	-46.7	33.2	-47.9	34.1	-49.4	35.3		
(5)			-43.7	21.4	2.0	1.2	-45.2	22.2	-46.4	22.9	-47.5	23.5	-49.0	24.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.9	-28.9	-22.7	-13.0	-1.5	8.1	14.9	18.4	16.0	8.6	-2.0	-17.3	-28.6	(6)	
	DBStd	17.62	5.80	5.35	5.82	5.02	4.12	3.54	2.91	3.19	4.26	5.36	7.62	6.29	(7)	
	HDD10.0	5738	1206	916	713	346	86	5	0	2	76	373	818	1197	(8)	
	HDD18.3	8181	1464	1149	972	596	317	113	37	84	293	631	1068	1456	(9)	
	CDD10.0	659	0	0	0	1	27	151	259	188	33	0	0	0	(10)	
	CDD18.3	59	0	0	0	0	1	9	38	12	0	0	0	0	(11)	
	CDH23.3	717	0	0	0	1	22	171	395	127	2	0	0	0	(12)	
	CDH26.7	164	0	0	0	0	2	36	108	18	0	0	0	0	(13)	
	WSAvg	1.4	0.7	1.0	1.6	1.9	2.0	1.6	1.4	1.3	1.5	1.6	1.1	0.7	(14)	
(15) Precipitation	PrecAvg	693	8	7	12	38	68	94	145	140	88	43	23	13	(15)	
	PrecMax	928	26	48	27	82	130	232	351	274	189	129	50	29	(16)	
	PrecMin	566	1	0	2	2	23	22	28	34	19	8	7	1	(17)	
	PrecStd	81	6	8	6	17	29	45	60	58	39	22	10	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.0	-4.9	5.2	17.7	26.4	29.8	31.8	28.7	23.0	15.3	0.3	-6.4	(19)	
		MCWB	-9.4	-6.6	0.3	7.7	13.5	18.1	19.3	18.9	15.1	8.2	-1.2	-7.6	(20)	
	2%	DB	-14.8	-7.9	1.8	13.1	22.8	27.5	29.6	26.5	20.3	10.4	-1.8	-12.8	(21)	
		MCWB	-15.5	-9.4	-1.9	5.8	11.7	16.7	18.5	18.0	13.8	5.3	-2.9	-13.4	(22)	
	5%	DB	-17.9	-10.8	-0.6	9.6	20.1	25.2	27.6	24.6	17.9	7.2	-3.8	-16.4	(23)	
		MCWB	-18.4	-12.1	-3.5	3.4	10.5	15.2	18.1	17.3	12.6	3.8	-4.7	-16.9	(24)	
	10%	DB	-20.5	-13.7	-2.9	6.8	17.3	23.0	25.5	22.5	15.6	4.9	-6.1	-19.1	(25)	
		MCWB	-20.9	-14.7	-5.2	2.1	9.1	14.8	17.8	16.6	11.4	2.2	-6.9	-19.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.5	-6.5	0.5	8.8	14.3	20.3	21.6	20.8	16.5	8.7	-0.5	-7.6	(27)	
		MCDB	-7.9	-4.9	4.4	15.7	23.5	26.1	26.8	26.1	20.5	13.5	0.3	-6.5	(28)	
	2%	WB	-15.4	-9.3	-1.5	6.3	12.8	18.4	20.2	19.5	14.8	6.3	-2.8	-13.4	(29)	
		MCDB	-14.8	-7.9	1.4	12.4	20.6	24.4	25.4	23.8	18.6	9.3	-1.9	-12.8	(30)	
	5%	WB	-18.4	-12.0	-3.3	4.0	11.3	17.1	19.4	18.6	13.6	4.3	-4.7	-16.9	(31)	
		MCDB	-18.0	-10.8	-0.7	8.7	17.8	22.0	24.3	22.2	16.8	6.8	-3.8	-16.5	(32)	
	10%	WB	-20.9	-14.6	-5.2	2.3	10.0	15.8	18.6	17.5	12.1	2.5	-6.8	-19.5	(33)	
		MCDB	-20.5	-13.6	-2.9	6.4	15.7	20.5	23.4	21.0	14.8	4.5	-6.1	-19.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	14.8	15.1	12.1	13.0	13.0	11.7	11.1	10.8	9.4	9.5	8.9	(35)	
		MCDBR	10.8	15.0	15.9	16.1	19.0	17.3	15.9	14.7	14.2	13.6	8.5	8.1	(36)	
		MCWBR	10.1	13.3	12.2	9.1	9.0	7.1	6.0	6.5	7.5	8.3	7.3	7.6	(37)	
	5% WB	MCDBR	10.7	14.7	14.8	15.0	16.1	13.4	12.3	11.4	12.0	12.4	8.2	8.0	(38)	
		MCWBR	10.0	13.0	11.4	8.7	8.2	6.3	5.8	6.0	6.9	7.9	7.2	7.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.249	0.285	0.347	0.474	0.526	0.489	0.475	0.412	0.373	0.376	0.299	0.249	(40)	
	taud		2.125	2.062	1.923	1.730	1.821	2.018	2.106	2.294	2.324	2.081	2.037	2.066	(41)	
	Ebn at Noon		727	803	813	744	731	770	773	809	796	673	622	635	(42)	
	Edh at Noon		68	102	146	204	197	163	147	115	99	101	75	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.35	2.52	4.18	5.25	4.98	5.53	5.04	4.22	3.12	1.96	1.34	1.01	(44)	
	RadStd		0.04	0.07	0.16	0.38	0.51	0.55	0.43	0.40	0.32	0.19	0.11	0.04	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.47	+0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page